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BR1601A02

Datasheet

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July 10, 2025

Version 1.5

Version History

Version	Revision	Date	Revised by
1.0	Initial version	2023-02-18	Yongqiang Xu
1.1	Update Bleutooth spec	2023-06-27	Sweet
1.2	Update MOQ	2023-12-22	Sweet
1.3	Update remark description	2024-04-24	Sweet
1.4	Review	2024-08-01	Sweet
1.5	Update RF parameter	2025-07-10	Yong Zheng

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Introduction

BR1601A02 is a Bluetooth SoC which supports BLE5.4. It includes a high performance and low power 32-bit RISC core which frequency is up to 64M, built-in 48K Bytes SRAM can be kept when the system is suspended，512K Bytes Flash can be used to store program which is high performance and low power.

BR1601A02 integrates an advanced BLE5.4 transceiver. It complies with Bluetooth BLE5.4 spec, and it can be configured as a standard 1Mbps BLE mode，enhanced, 125kbps BLE long range mode(S8) and 500kbps BLE long range(S2)。It supports AoA and AoD feature，RSSI，Master/Slave all-in-one, multiplex links，extended data packet length, KEYSCAN, IRC,10 bit 1.33Msps generic ADC (can be configurable to 16bit 16Ksps), analog MIC input, PGA, basic timer, generic timer, advanced timer, RTC, WWDG, IWDG, LPUART, USART, SPI, I2C, etc.

Processor

- 32bit RISC, frequency up to 64MHz
- Support upgrade by SWD interface
- Support debug mode

Power management

- Power supply range: 1.8V~3.6V
- Embedded power regulator, DCDC and LDO
- Power detection
- Receiver current: 3.8mA@3.3V
- Transmitter current：4.2mA @0dBm/3.3V
- Sleep mode (48KB RAM keep): 1.6μA@3V
- Power off mode (SOC wake-up by wake up and reset PINs): 130nA

Software

- Completely comply with Bluetooth BLE5.4 specification with optimizing power consumption of Stack and Bluetooth controller
- Support Bluetooth Application profiles and Bluetooth Peripheral program
- Support OTA upgrade

Memory

- 48K Bytes SRAM
- 512K Bytes FLASH

Clock

- HSE：32MHz external high-speed crystal
- LSE：32.768KHz external low-speed crystal
- HSI：64MHz internal high-speed RC oscillator
- LSI：32KHz internal low-speed RC oscillator
- Supports 1 clock output, can be configured to different clocks, 4 frequency division output.

Peripheral

- One 5-channel DMA controller
- Two USART interface, up to 8 Mbps (can be configured as ISO7816, IrDA, LIN)
- One LPUART interface, support low power feature，up to 9600bps under Sleep mode, support wake-up under Sleep mode.

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- 2 SPI interface，up to 16MHz，can be configured as Master or Slave role and can be configured as I2S interface.
- One I2C interface，up to 1MHz, can be configured as Master or Slave role.
- 21 GPIO with IO-MUX support
- Support Power on Reset, Power off Reset and external PIN reset
- Support Watch Dog reset
- One 16bit advanced timer, support input capture, output comparison, PWM output and quadrature encoding signal decoding and other functions；4 independent channels, 3 of which support 6 complementary PWM outputs.
- One 16-bit universal timing counter, support input capture, output comparison, PWM output, mono-pulse output, 4 independent channels
- One 16-bit basic timer
- One 24-bit system timer
- One 7-bit window Watch Dog (WWDG)
- One 12-bit independent Watch Dog (IWDG)
- One 10-bit 1.33Msps generic ADC，support five external single-ended channels and two internal channels
- Built-in PGA supports up to 128x amplification.
- MIC BIAS voltage is adjustable from 1.6V to 2.3V
- One IR transmitting controller with configurable infrared remote-control protocol.
- KEYSCAN module, support 8/10/13 GPIO can be configured with 44/65/104 button function.
- RTC real-time clock, support leap year perpetual calendar, alarm events, periodic wake up
- Supports hardware CRC16 and CRC32 operations.

RF characteristics

- BLE 1Mbps receive sensitivity：-96dBm
- BLE 2Mbps receive sensitivity：-93dBm
- Programmable transmit power, up to +8dBm
- Single-ended antenna

Operating condition

- Operating temperature range：-40°C~85°C
- ESD：±2KV(Human body model)

Package

- QFN32(4mm x 4mm)

Applications

- Smart Home and building automation.
- Connect appliances, lighting, locks, gateways, security system.
- Industry
- Logistics, production and manufacturing, automation, asset tracking and management, remote display, cable replacement, human machine interface, access control
- Retail
- Beacons, advertising, electronic shelf labels and price tags, point of sale and payment system.
- Health and medical care
- Thermometer, blood glucose and pressure gauge, weight scale, life monitoring, hearing aid
- Sports and fitness
- Collect exercise monitors and fitness trackers, heart rate monitors, running sensors, bike sensors, sports watches, gym equipment, team sports equipment.
- Remote control, keyboard and mouse, gaming peripherals
- Accessory
- Toys, trackers, luggage tags, wearables

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1. Chip Diagram

BR1601A02 integrates power management, clock circuit, Bluetooth baseband, modem and RF circuit. Microprocessor (CPU) controls Bluetooth baseband, SRAM, DMA, FLASH, ADC, CRC, IRC via AHB bus. Other peripheral CPUs are controlled by AHB to APB bridge.

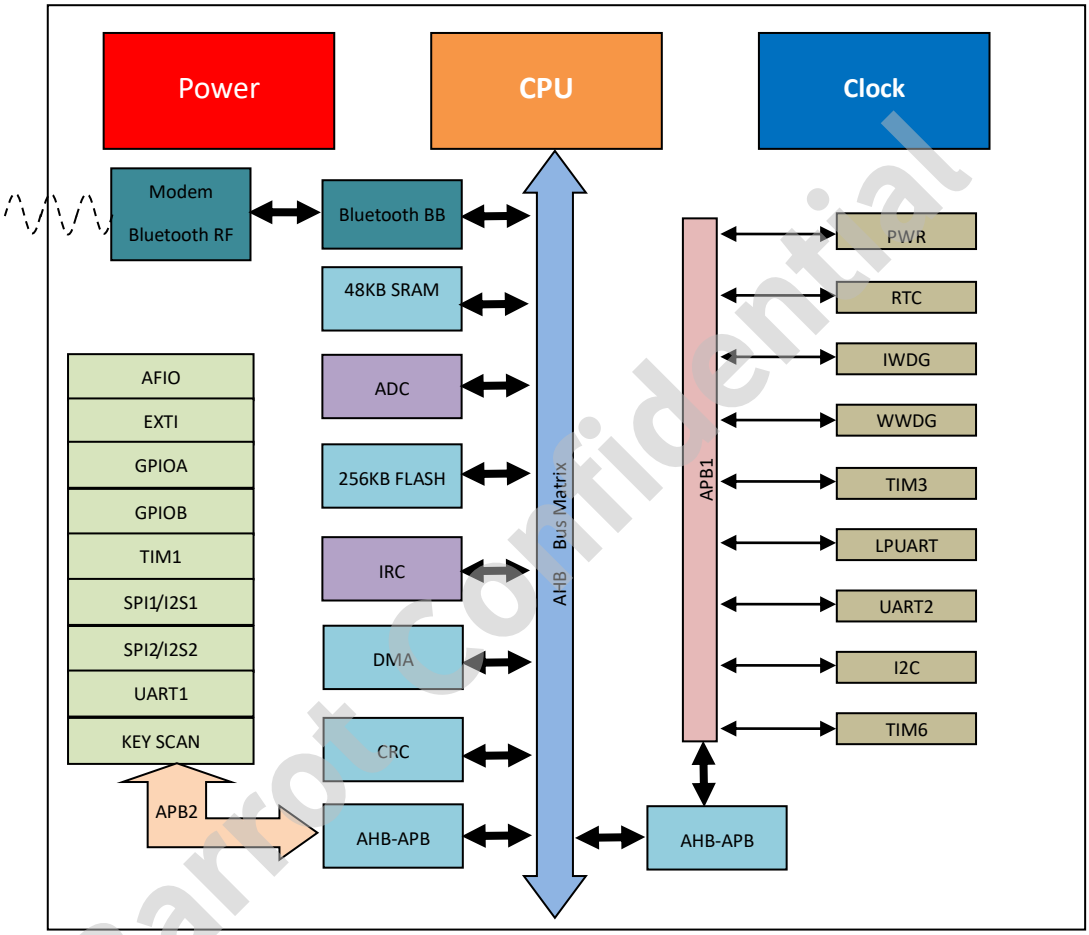


Figure1 BR1601A02 Diagram

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2. Pin map

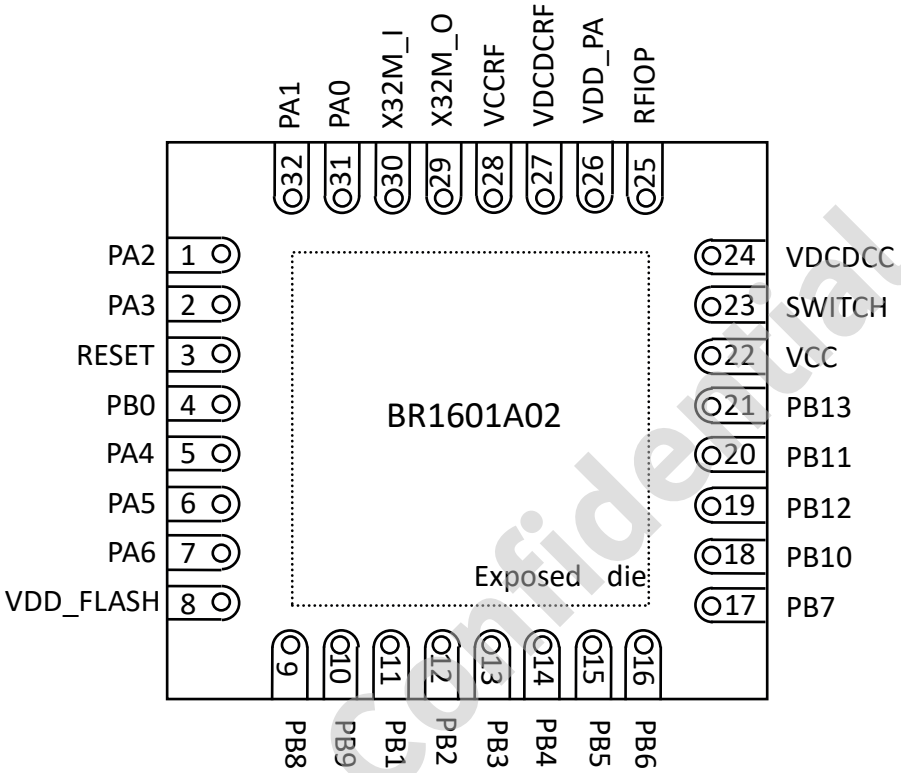


Figure 2 BR1601A02 Top view

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3. PIN definition

3.1. PIN description

Table1 BR1601A02 PIN symbol description

Symbol	Description
I	Input
O	Output

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I/O	Bidirectional
OD	Out drain
P	Power
G	Ground
AIO	Analog IO

Table 2 BR1601A02 PIN description

PIN Name (default function)	No.	IO type	MUX	Function description
PA2	1	I/O	SPI1_MOSI(I2S1_SD) KEY3	-
PA3	2	I/O	SPI1_MISO(I2S1_MCK) KEY4	-
RESET	3	AIO	-	-
PB0	4	I/O	TIM1_CH1 SPI2_NSS(I2S2_WS) USART1_RTS LPUART_RTS KEY11	-
PA4	5	I/O	TIM1_CH3N	-

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(SWDCLK)			USART1_TXD(7816_TX1) KEY9	
PA5 (SWDIO)	6	I/O	TIM1_ETR USART1_RXD(7816_RST1) KEY10	-
PA6	7	I/O	TIM1_BKIN USART2_TXD USART1_CK(7816_CLK1) KEY5	-
VDD_FLASH	8	P	-	External 2.2uF capacitor
PB8 (X32K_I)	9	I/O	TIM1_CH1 IIC_SDA USART1_RTS KEY7	ADC3
PB9 (X32K_O)	10	I/O	TIM1_CH2 IIC_SCL USART1_CTS KEY8	ADC2
PB1	11	I/O	TIM1_CH2 SPI2_CLK(I2S2_CLK)	-

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			USART1_CTS LPUART_TXD KEY12 ANT_SW4	
PB2	12	I/O	TIM1_CH3 SPI2_MOSI LPUART_RXD KEY13 ANT_SW5	-
PB3	13	I/O	TIM1_CH4 SPI2_MISO(I2S_MCK) LPUART_CTS PA_LDO_EN ANT_SW1	WAKEUP
PB4	14	I/O	SPI2_CLK(I2S2_CLK) TIM3_CH1(IRC_TX) USART2_TXD(7816_TX2) ANT_SW6	-
PB5	15	I/O	SPI2_MISO(I2S2_MCK) TIM3_CH2(IRC_RX)	-

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			USART2_RXD(7816_RST2) RCC_MCO ANT_SW7	
PB6	16	I/O	SPI2_MOSI(I2S2_SD) TIM3_CH3 IIC_SDA USART1_TXD SWDCLK ANT_SW2	ADC5
PB7	17	I/O	SPI2_NSS(I2S2_WS) TIM3_CH4 IIC_SCL USART1_RXD SWDIO ANT_SW3	ADC4
PB10	18	I/O	TIM1_CH3 LPUART_RTS USART2_RXD IIC_SMBA KEY6	ADC1

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PB12	19	I/O	TIM1_CH1N LPUART_TXD USART2_CTS	AMIC_BIAS
PB11	20	I/O	TIM1_CH4 LPUART_RXD USART2_RTS IIC_SMBA	AMIC_N
PB13	21	I/O	TIM1_CH2N LPUART_CTS USART2_CK	AMIC_P
VCC	22	P	-	Chip power supply
SWITCH	23	P	-	external DCDC
VDCDC	24	P	-	DCDC output
RFIOP	25	AIO	-	Antenna interface
VDD_PA	26	P	-	PA power supply
VDCDCRF	27	P	-	DCDC output, connect to VDCDC
VCCRF	28	P	-	Chip power supply
X32M_O	29	AIO	GPIO31	External 32MHz crystal
X32M_I	30	AIO	GPIO32	External 32MHz crystal

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PA0	31	I/O	SPI1_NSS(I2S1_WS) KEY1	-
PA1	32	I/O	SPI1_CLK(I2S1_CLK) KEY 2	-
Ground PAD	---	P	-	Ground

Note：after reset，I/O PINs will reside in Analog input mode, except the following PINs:

- RESET：pull-high input mode
- PA4：SWCLK pull-low input mode.
- PA5：SWDIO pull-high input mode.

4. Function description

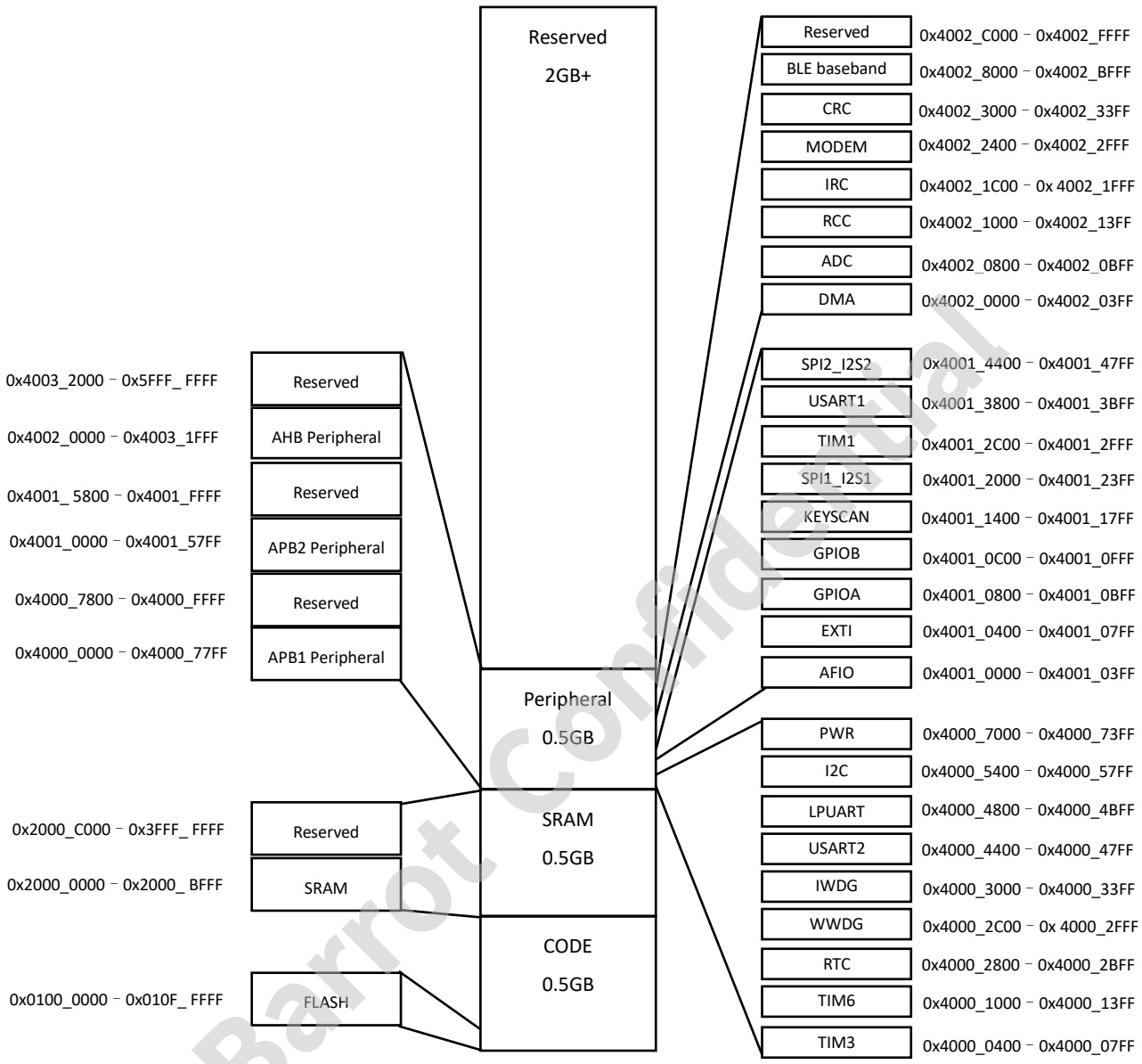
4.1. Processor core

BR1601A02 integrates a 32-bit RISC kernel. The processor runs the Bluetooth protocol stack and all user applications.

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4.2. Memory



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Figure 3 BR1601A02 memory map

4.2.1. FLASH

512K Bytes Flash, all 256K-bytes memory can be used for program storing.

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4.2.2. SRAM

48KBytes SRAM，all can be kept under Sleep mode.

4.3. Serial peripheral interface (SPI)

The chip provides two SPI interfaces, which support half-duplex or full-duplex communication. Users can configure it in master mode, multi-master mode, and slave mode. In addition, the interface provides CRC check function to ensure reliable communication.

SPI main functions include:

- Full-duplex synchronous transmission
- Frame format: 8-bit, 16-bit
- Master or Slave role
- Multi Master mode
- Fast communication under Master mode and Slave mode
- NSS supports software or hardware control in both master mode and slave mode: the master/slave mode of operation can be dynamically changed
- Clock polarity and phase are programmable
- Programmable data order, MSB first or LSB first
- Dedicated send and receive flags that trigger interrupts
- SPI bus busy status flag
- Hardware CRC
  - In send mode, the CRC value can be sent as the last byte
  - Performs the CRC check on the last received byte automatically in full-duplex mode
- Main-mode failure, overload, and CRC error flags trigger interrupts
- Single byte send and receive buffers that support DMA: generate send and receive requests
- Highest baud rate：16Mbps

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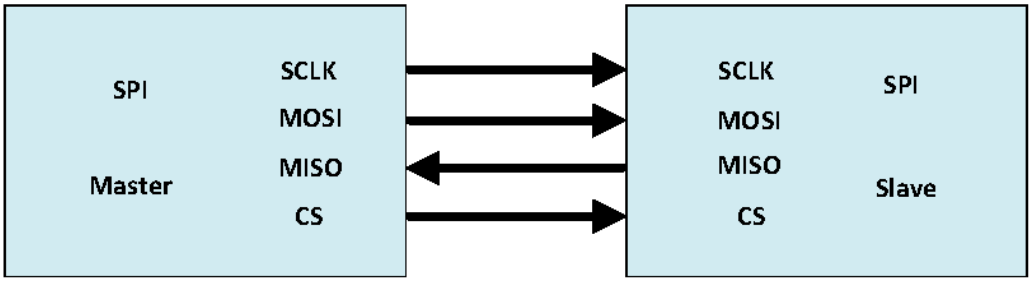


Figure 4 SPI connection

4.4. 4.4. Universal synchronous asynchronous transceiver (USART)

The chip integrates three serial transceiver interfaces, including two universal synchronous/asynchronous transceivers (USART1, USART2) and one universal asynchronous transceiver (LPUART), which support low power consumption. USART1 and USART2 support synchronous/asynchronous communication, and support IrDA SIR ENDEC transmission codec, multi-processor communication mode, single-line half-duplex communication mode and LIN master/slave function.

USART1 and USART2 interfaces have hardware CTS and RTS signal management capabilities, while also compatible with ISO7816 smart card mode. In addition, all interfaces support DMA operations.

USART main features are:

- Full duplex asynchronous communication
- Single-wire half-duplex communication
- NRZ standard format
- Programmable baud rate, up to 4Mbits/s, support fractional baud rate
- Programmable data length (8-bit or 9-bit)
- Stop bit can be configured as 1 or 2
- LIN master mode can send synchronous disconnections and LIN slave mode can detect disconnections. When USART hardware is configured as LIN, 13-bit disconnections symbol can be generated, and 10/11-bit disconnection symbol can be detected
- The output clock is used for synchronous transmission.

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- IRDA SIR Encoder decoder, in normal mode, supports a duration of 3/16 bits.
- Smart card simulation
  - Supports the asynchronous smart card protocol, ISO7816-3
  - Supports 0.5 and 1.5 stop bits
- Configurable DMA multi-buffer communication using DMA buffering to receive/send bytes in SRAM
- Independent bit for transmitter/receiver enabling
- Status flag
  - Receiver FIFO full
  - Transmitter FIFO empty
  - Transmit end flag
- CRC
  - Transmit CRC
  - Check received data
- Error detection flag
  - Overflow error
  - Noise error
  - Frame error
  - CRC error
- 10 USART interrupts
  - CTS change
  - LIN disconnection symbol detection
  - Transmit data register empty.
  - Transmit end
  - Receive data register full
  - Bus idle
  - Overflow error

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- Frame error
- Noise error
- CRC error

- For multiprocessor communication, silent mode is entered if the addresses do not match
- Wake up from silent mode (by detecting bus idle or detecting address flags)
- Two ways to wake up the receiver: address bit (MSB, bit 9), bus idle
- Mode Configuration:

Table 3 BR1601A02 USART function list

Communication mode	USART1	USART2	LPUART
Asynchronous mode	Y	Y	Y
Hardware flow control	Y	Y	Y
Multiple cache communication (DMA)	Y	Y	Y
Multiple processor communication	Y	Y	N
Synchronous mode	Y	Y	N
Half-duplex mode (Singe wire mode)	Y	Y	N
Smart card	Y	Y	N
IrDA	Y	Y	N

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LIN	Y	Y	N

4.5. Inter-IC Sound (I2S)

I2S is a digital audio transmission standard that uses 3 pins for synchronous serial communication and works in master-slave mode. When transmitting audio data, it supports 16 -bit, 24 -bit, or 32-bit data widths and can be configured as an input or output channel. In addition, I2S supports four audio standards, including the Philips I2S standard, the MSB and LSB Alignment standard, and the PCM standard. The I2S also supports a wide range of audio sampling frequencies, ranging from 8kHz to 96kHz. The standard has been widely used in the connection between audio processors, digital signal processors and audio decoders, as well as audio equipment, smart home, car audio and other fields.

I2S supports half-duplex communication and can work in both master and slave modes. When acting as a master device, it provides a clock signal to an external slave device.

I2S interface function are:

- Half-duplex communication (Either transmit or receive at one time)
- Support Master and Slave mode
- 8-bit programmable pre-divider which can generate accurate audio 8KHz~96KHz sampling frequency
- Data format (16-bit, 24-bit, 32-bit)
- Audio channel fixed packet frame length is either 16-bit(16 bit data frame) or 32-bit (16, 24 or 32 bit data frame)
- Programmable clock polarity (steady state)
- Underrun flag in Transmit mode and Overflow flag in Master/Slave mode
- 16-bit data register is used for sending and receiving, with one register at each end of the channel
- Support I2S protocols

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- I2S Philips standard
- MSB align (left align)
- LSB align (right align)
- PCM standard (16-bit channel frames with long or short frame synchronization or 16-bit data frames extended to 32-bit channel frames)

- Data direction is always MSB first
- Both transmitting and receiving support DMA

4.6. I2C bus interface(I2C)

I2C bus is a two-wire communication interface, which supports multi-host functions and controls I2C bus timing, protocol, arbitration and timing. Supports multiple communication rates (up to 1MHz), supports DMA, and is compatible with SMBus 2.0. The I2C unit has a variety of uses, including CRC code generation and verification, SMBus(System Management Bus) and PMBus(Power Management Bus).

I2C functions are:

- Multi-host function: can be a master device or a slave device
- I2C master device
  - Generate clock
  - Generate Start and Stop signal
- I2C slave device
  - Programmable address detection
  - Support 7-bit or 10-bit addressing, support dual slave address responses in 7-bit slave mode
  - Stop bit check
- Generate and detect 7-bit / 10-bit address and broadcasting
- Support different communication speed
  - Standard (up to 100kHz)
  - Fast (up to 400kHz)

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  - Fast+(up to 1MHz)
- Support multiple status flags
  - Transmitter/Receiver mode flag
  - Byte transmit end flag
  - I2C bus busy flag
- Support multiple error flags
  - Arbitration loss in Master mode
  - ACK error occurred after address/data transmission.
  - Wrong start or stop condition detected.
  - Overflow or underrun when disabling stretch clock.
- One interrupt vector
  - Event interrupt and error interrupt share one same interrupt vector
- Optional clock stretch function
- Support DMA for single byte cache.
- Generation or verification of configurable PEC(Packet error detection)
  - PEC value in send mode can be transmitted as the last byte
  - Receives the last byte of the PEC error check in receive mode
- Compatible with SMBus 2.0
  - 25ms clock low timeout delay
  - 10ms master device cumulative clock low expansion time
  - 25ms low expansion time from device cumulative clock
  - Hardware PEC generation/verification with ACK control
  - Supports Address Resolution Protocol (ARP)
- Compatible with SMBus

4.7. General Input/Output (GPIO)

GPIO(General purpose input/output) is general I/O. AFIO(Alternate-function input/output) is

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alternate function I/O. The chip supports up to 21 GPIOs, which are divided into two groups (GPIOA and GPIOB), where group A has 7 ports and Group B has 14 ports. These GPIOs share pins with other peripherals, allowing users to flexibly configure them. Each GPIO can be configured independently as an output, input, or reusable peripheral function port, and in addition, they have a high current passing capability

GPIO main features are:

- GPIOs can be configured into following modes by software:
  - Floating input
  - Pull-high input
  - Pull-low input
  - Analog function
  - Open drain output
  - Push-pull output
  - Push-pull multiplex
  - Open drain multiplex
- Separate bit set or clear function
- All IOs support external interrupt
- All IOs support low power mode wakeup, interrupt rising or fall edge can be configured
- 8 EXTI can be used for wakeup from sleep, all IOs can be multiplexed as EXTI
- PB3 can be used as wakeup IO in PowerDown mode, rising edge as interrupt trigger
- Supports I/O multiplex by software remapping
- support GPIO lock mechanism. The lock status can be cleared by resetting
- Each I/O port register bit can be arbitrarily programmed, but must be accessed as a 32-bit word(16-bit half-word or 8-bit byte access is not allowed)

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4.8. KEYSCAN

Support 8/10/13 I/O ports, corresponding to 44/65/104 buttons.

Button shake elimination can be configured.

The chip supports three scanning modes: automatic scanning, software scanning, and low power consumption.

- Automatic mode: can be configured to automatically scan at a fixed interval.
- Low power mode: Can be configured to detect when the key is pressed to start a total of three scans.
- Software Mode: Scan mode triggered by software.

4.9. Analog/digital converter (ADC)

BR1601A02 supports 10-bit 1.33Msps ADC, supports single-ended or differential AMIC, and has a built-in PGA with a maximum gain of 42dB.

The chip supports a 10-bit 1.33Msps generic ADC. It supports single-end or differential AMIC. It has embedded PGA. The gain is up to 42dB.

The chip provides voltage adjustable MICBIAS supply power to MIC, output 1.6V~2.3V optional. It supports up to 8 channels, 5 external single-ended, 1 differential MIC and 2 internal channels.The internal channel includes a VCC detection channel and a temperature sensor channel. 5 external channels, channel 1(PB10), 2(PB9) detection range 0V-1V, channel 3(PB8), 4(PB7), 5(PB6) detection range 0V-3.6V, in where channel 3(PB8), 4(PB7), 5(PB6) input voltage <=VCC+300mV.

In Audio mode, it utilizes a built-in programmable gain amplifier (PGA) and microphone bias. It amplifies the MIC signal through the PGA, then converts to digital signals through the analog ADC. After audio input control (low-pass decimation filtering and selectable energy and zero-crossing detection), the audio data is stored into the system RAM via DMA, and finally outputs 16-bit 16kHz audio signal format data.

ADC main features are:

- Support AMIC input, adjustable MIC bias
- PGA support single end or differentiated input，adjustable gain value

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- Support 1 ADC, 5 measurable external single-ended, 1 differential MIC and 2 internal channels, optional input channels
- Internal channel support TempSensor、VCC
- 10bit 1.33Msps ADC
- Digital extraction filtered to 16 bits，support digital filter noise filtering
- Support single and continuous conversion modes
- Channel conversion can be accompanied by DMA request generation
- Analog watchdog feature allows applications to detect if the input PB10 voltage exceeds a user-defined high/low threshold
- Interruption occurs when the conversion ends and a simulated watchdog event occurs
- In audio mode, filter output data is stored in 16bit data register; In general mode, data is stored in 16bit data register in LSB align
- In audio mode, it outputs in 16bit 16Ksps signed mono PCM format; in general mode, it outputs in 10bit 1.33Msps unsigned format.

4.10. Low power mode

BR1601A02 supports 4 low power mode

- Idle mode

when CPU stops running, all peripherals are working and interrupts/events can occur to wake up the CPU.

- Standby mode

Power supply is running normally, CORE power domain powers off, and BLE can be operated.

- Sleep mode

High-speed clock is turned off, power supply is running in low-power mode, CORE power domain powers off, and BLE powers off.

- PD mode

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All systems turn off, and CPU can be woken up by WAKEUP IO and RESET IO.

4.11. External Interrupt/Event controller (EXTI)

The chip's external interrupt/event controller contains a total of 14 edge detection circuits to generate interrupt/event requests. Each break line can be configured independently as an event or break, and supports 3 trigger types: rising edge, falling edge, or double . At the same time, each break line can also be shielded independently. The device's suspend register maintains the interrupt request for the status line and clears the corresponding interrupt request by writing '1' to the corresponding bit of the suspend register.

4.12. Direct Memory Access (DMA)

This chip has a built-in general-purpose 5-channel DMA controller, which can manage memory-to-memory, peripheral-to-memory, and memory-to-peripheral data transfer. Each channel comes with a dedicated hardware DMA request logic and can be triggered by software. In addition, the transmission length, source and destination address of each channel can be set individually by software.
The DMA controller supports peripherals, including SPI, I2S, I2C, USART, as well as general-purpose, basic, and advanced control timers TIMx and ADCs.

4.13. Cyclic Redundancy Check Computing Unit (CRC)

The chip integrates CRC32 and CRC16 circuit units, and the cyclic redundancy check (CRC) circuit unit obtains the CRC calculation result according to the fixed generation polynomial.

In many applications, CRC-based technologies are used to verify the consistency of data transmission or storage. Within the scope of the EN/IEC 60335-1 standard, it provides a method of detecting flash memory errors, CRC computing units can be used to calculate the signature of software in real time and compare it with the signature generated when linking and generating the software.

CRC main features are:

- CRC16: Polynomial X16 + X15 + X2 + X0
- CRC16 Calculation Time: 1 AHB Clock Cycle (HCLK)

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- CRC32：Polynomial X32+X26+X23+X22+X16+X12+X11+X10+X8+X7+X5+X4+X2+X+1
- CRC32 Calculation Time：4 个 AHB Clock cycle(HCLK)
- Configurable cyclic redundancy calculation initial value
- Support DMA

4.14. Clock system

2 high-speed clocks:

- HSI oscillator clock (64MHz)
- HSE oscillator clock (32MHz)

2 secondary clock sources:

- LSI oscillator clock (32KHz)
- LSE oscillator clock (32.768KHz)

After the system powers on or reset, the HSI and HSE are enabled by default. The system clock is HSI by default. LSI can be used to drive the standalone watchdog IWDG and configured to drive RTC, KEYSCAN, and LPUART. LSE can be configured to drive RTC, KEYSCAN, and LPUART. LSI/LSE can be used to wake up the system from Idle/Standby/Sleep/PowerDown mode. When not in use, either clock source can be turned on or off independently to optimize system power consumption.

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4.15. Real-Time Clock (RTC)

The real-time clock (RTC) has a set of independent BCD timers/counters that count continuously. Under the corresponding software configuration, the function of the calendar is available. At the same time, the RTC is programmed to provide an alarm interrupt.

Two 32-bit registers record time in decimal format (BCD), including sub-seconds, seconds, minutes, hours (in 12 or 24-hour format), days (days of the week), days (days), months, and

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years.

A separate 32-bit register provides sub-second values in binary format. The additional 32-bit registers provide programmable seconds, minutes, hours, days, days, months, and years.

The RTC provides the funcationability to wake up the system in low-power mode.

4.16. Infrared Controller (IRC)

The IR controller can be configured to different encoding formats by software in order to implement different IR protocol signals and it s,support the software IR self-learning function.

IRC main features are:

- Carrier frequency range: 30kHz~60kHz
- Support pulse width coding and pulse distance coding
- Support Manchester encoding
- Support No carrier mode
- Support any combination of mark code and space code
- Support 16x21-bit wide Code FIFO to store encoding commands
- Support command repetition
- Generate interrupt when transmission finishes

4.17. Serial Wire Debug (SWD)

The chip has a built-in SWJ-DP interface. This is a combination of JTAG and serial single-line debugging interface. It can realize the serial single-line debugging interface or JTAG interface connection. JTAG's TMS and TCK signals respectively share pins with SWDIO and SWCLK.

4.18. Timer and watchdog

The chip has an advanced control timer, a normal timer, a basic timer, two watchdog timers and a system tick timer.

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4.18.1. Basic timer (TIM6)

The basic timer (TIM6) contains a 16-bit autoload counter driven by a programmable prescaler. It provides time baseline for generic timer.

Basic timer main functions are:

- 16bit autoload accumulation counter.
- 16bit programmable (real-time modifiable) prescaler, supporting frequency division of the input clock by any numerical value with a coefficient between 1~65536.
- Generate interrupts/DMA requests when events update (counter overflow).

4.18.2. Generic Timer (TIM3)

The generic timer has a 16-bit autoload increment/decrement counter, a 16bit prescaler and 4 independent channels. Each channel can be used for input capture (for measuring pulse width), output comparison, PWM, and single pulse mode output.

Generic timer main functions are:

- 16-bit up, down, up/down auto-load counters
- 16-bit programmable (can be modified in real time) prescaler, the division coefficient of the counter clock frequency is any value between 1~65536
- 4 independent channels:
  - Input capture
  - Output comparison
  - PWM generation (edge or middle align mode)
  - Single-pulse mode output
- Synchronous circuits that use external signals to control timers and timer interconnections
- Interrupts/DMAs occur when the following events occur:
  - Update Events: Counter Overflow Up/Down, Counter Initialization (Triggered by Software or Internally)
  - Trigger events (counter start, stop, initialize, or count triggered internally)

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- Input capture
- Output comparison

- Support incremental (quadrature) encoders and Hall sensor circuitry used in positioning applications
- The trigger input can be used as an external clock or current management by cycle

4.18.3. Advanced control timer (TIM1)

The advanced control timer consists of a 16bit autoload counter driven by a programmable prescaler. It supports a variety of functions, including measuring the pulse width of the input signal (input capture) or generating an output waveform (output comparison, PWM, complementary PWM output with embedded dead time, etc.). By using a timer prescaler and an RCC clock control prescaler, the pulse width and waveform period can be adjusted, ranging from a few microseconds to a few milliseconds.

Advanced timer main functions are:

- 16bit up, down, up/down autoload counters
- 16bit programmable prescaler that can be modified in real time, and the crossover coefficient is any value between 1~65536
- Support 64MHz as a maximum input clock
- 4 independent channels:
  - Input capture
  - Output comparison
  - PWM generation (edge or middle align mode))
  - Single-pulse mode output
- Trigger time is configurable by software over the full PWM cycle
- Dead-time programmable complementary output is configurable
- An external signal is used to control the timer, and the synchronization circuit of multiple timers
- Update the repeat counter of the timer register after a specified number of counter

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cycles finishes

- The Break input signal can put the timer output signal in a reset status or a known status
- Following events that can trigger interrupts:
  - Update: Counter overflow up/down, counter initialization (via software or internal/external triggering)
  - Trigger events (counter start, stop, initialize, or count triggered internally/externally)
  - Input capture
  - Output comparison
  - Break signal input
- Support incremental (quadrature) encoders and Hall sensor circuitry used in positioning applications
- The trigger input can be used as an external clock or current management by cycle

In debug mode, the counter can be frozen while the PWM outputs are disabled, thus cutting off the switches controlled by these outputs. Many of the functions are same with the standard TIM timers, and the internal structure is also the same, so advanced control timers can operate in conjunction with TIM timers through the timer link function to provide synchronization or event link functions.

4.18.4. System Time-Based Timer (STICK)

This timer can be used either as a dedicated timer for real-time operating systems or as a standard decrement counter. It has the following characteristics:

- 24bit decrement counter
- Automatic re-load
- When the counter is 0, a maskable system interrupt can be generated
- Programmable clock source

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4.18.5. Watchdog timer (WDG)

The chip contains two watchdogs, an independent watchdog (IWDG) and a window watchdog (WWDG). Two watchdogs provide enhanced security, timing accuracy and use flexibility.

Independent watchdog (IWDG)

The independent watchdog consists of a 12bit decrement counter and a 8bit prescaler, driven by a separate low-speed RC oscillator that remains valid even if the master clock fails. Once IWDG is activated, if the watchdog counter is not cleared within the set time, a reset is generated when the counter counts to 0x000.
It can be used to reset the entire system when the application fails, or work as a free timer to provide timeout management for application.

Window Watchdog (WWDG)

Window watchdogs are often used to monitor software failures caused by external interference or unforeseen logical conditions where an application deviates from the normal sequence of operations. If the decrement counter value is not refreshed before the T6 bit becomes 0, the watchdog circuit will produce a chip reset when the preset time period is reached. If the 7bit decrement counter value (in the control register) is refreshed before the decrement counter reaches the window register value, a chip reset will also be generated. This indicates that the decrement counter needs to be refreshed in a limited time window.

Main functions are:

- WWDG is divided by APB1 clock and used as the clock input
- Programmable decrement counter
- Conditional reset
- If the watchdog is activated, a reset will be generated when the decrement counter value is less than 0x40
- If the watchdog is activated, a reset is generated when the decrement counter is reloaded outside the window

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- If the watchdog is started and interrupts are allowed, an Early Wake Interrupt (EWI) is generated when the counter is equal to 0x40, which is used to reload the counter to avoid WWDG resets.

5. Electrical characteristics

5.1. Test conditions

Unless otherwise specified, all voltages are based on Vss.

5.1.1. Minimum and maximum values

Unless specified, all minimum and maximum values are guaranteed at the worst ambient temperature, supply voltage and clock frequency conditions by testing 100% of the products on the production line at ambient temperature TA = 25°C and TA = TAmax (TAmax matches the selected temperature range).
The notes at the bottom of each table indicate that the data obtained through laboratory tests, design simulations and/or process characteristics are not tested on the production line, and on the basis of laboratory tests, the minimum and maximum values are obtained by adding or subtracting the average of the samples (mean ± 3Σ).

5.1.2. Typical values

Unless specified, typical data are based on TA = 25°C and VCC = 3.3V (1.8V≤VCC≤3.6V voltage range). This data is for design guidance only and has not been tested.

5.1.3. Typical curves

Unless specified, typical curves are for design guidance only and have not been tested.

5.1.4. load capacitor

The load conditions when measuring pin parameters are shown in the figure below.

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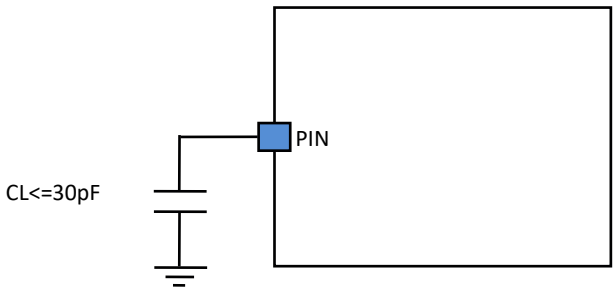


Figure 5 Load capacitor test

5.1.5. PIN input voltage

The measurement method of PIN input voltage is shown in the figure below.

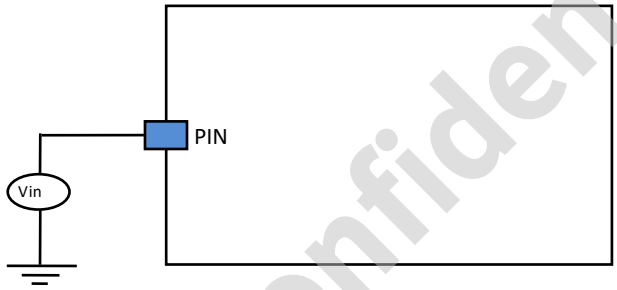


Figure 6 PIN input voltage test

5.1.6. Power supply solution

Please refer to the BR1601A02 schematics.

5.1.7. Power consumption measurement

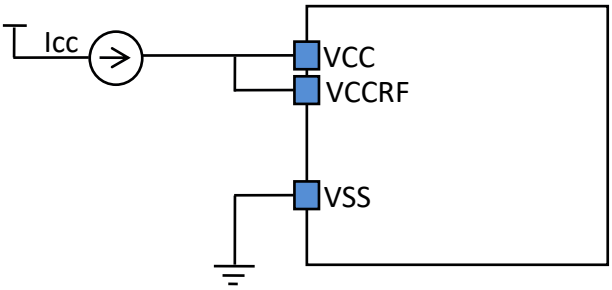


Figure 7 Power consumption measurement

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5.2. Absolute Maximum Ratings

Loads placed on components more than values of "absolute maximum rating" given in the table below may result in permanent damage to components. This only gives the maximum load that can be sustained and does not mean that the functional operation of components under these conditions is working properly. Long-term operation of components at maximum conditions will affect the reliability of the device.

Table 4 voltage characteristics

Symbol	Description	Min.	Max.	Unit
VCC-VSS	External main supply VCC(1)	-0.3	3.6	V
V_{IN}	Input voltage on other PINs(2)	VSS-0.3	VCC+0.3	V
V ESD(HBM)	ESD Discharge Voltage (Human Body Model)	Refer to the Absolute Maximum (Electrical Susceptibility) section		

(1) All power supply (VCC) and ground (VSS) pins must always be connected to the power supply system within the allowable external range.

(2) The IINJ (PIN) must never exceed its limit (see the table below), i.e. it is guaranteed that the VIN does not exceed its maximum value. If it is not possible to guarantee that the VIN does not exceed its maximum value, it is also necessary to ensure that the external restriction IINJ (PIN) does not exceed its maximum value. When VIN <VSS, there is a reverse injection current.

Table 5 Current characteristics

Symbol	Description	Maximum	Unit
IVCC	Total current through VCC/VCCRF power line (supply current) (1)	150	mA

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IVSS	Total current through VSS ground (outgoing current)(1)	150	
IIO	Output sink current on arbitrary I/O and control pins	12	
	Output current on arbitrary I/O and control pins	-12	
$I_{INJ(PIN)}$ (2)(3)	Injection current at the RESET pin	±5	
	Injection current for the OSC_IN pin of the HSE and OSC_IN pin of the LSE	±5	
	Injection current for other pins (4)	±12	
$\Sigma I_{INJ(PIN)}$ (2)	The total injection current on all I/O and control pins (4)	±150	

(1) All power (VCC, VCCRF) and ground (VSS) pins must always be connected to the power supply system within the allowable external range

(2) The IINJ (PIN) must never exceed its limit, i.e. it is guaranteed that the VIN does not exceed its maximum value. If it is not possible to guarantee that the VIN does not exceed its maximum value, it is also necessary to ensure that the external restriction of the IINJ (PIN) does not exceed its maximum value

(3) When several I/O ports have injection currents at the same time, the maximum value of the Σ IINJ (PIN) is the sum of the immediate absolute values of the forward and reverse injection currents

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Table 6 Temperature characteristics

Symbol	Description	Value	Unit
T_{STG}	Storage temperature range	-40 ~ + 125	°C
T_J	Maximum junction temperature	105	°C

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5.3. Work conditions

5.3.1. Generic work conditions

Table 7 Generic work conditions

Symbol	Parameter	Min.	Max.	Unit
f HCLK	Internal AHB clock frequency	-	64	MHz
f PCLK1	Internal APB1 clock frequency	-	32	
f PCLK2	Internal APB2 clock frequency	-	64	
VCC	Standard work voltage	1.8	3.6	V
VCCRF	Analog work voltage	1.8	3.6	V
T_A	Environment temperature	-40	85	°C
T_J	Conjunction temperature range	-40	105	°C

5.3.2. Work conditions on Power on and Power down

The parameters given in the table below are based on the ambient temperatures listed in the table below.

Table 8 Work conditions on Power on and Power down

Symbol	Parameter	Condition	Min.	Max.	Unit
T_{VCC}	VCC rise rate	VCC=3.3V	20	∞	us/V
	VCC fall rate		100	∞	

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5.3.3. Embedded reset and power control module features

The parameters given in the table below are based on the ambient temperature and VCC supply voltage listed in the table (General Operating Conditions).

Table 9 Embedded reset and power control module features (1)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
V_{BOR}	VCC power-on threshold voltage	T_A =25°C	-	1.65(1)	-	V
	VCC power down threshold voltage	T_A =25°C	-	1.60(1)	-	
$V_{BORhyst}$	BOR hysteresis	T_A =25°C	-	20	-	mV

(1) Guaranteed by design, not tested in production.

5.3.4. DCDC characteristics

DCDC is an internal voltage generating module, and the parameters given in the table below are based on the ambient temperature and VCC supply voltage listed in the table (General Operating Conditions).

Table 10 Characteristics of the Embedded DCDC power management module(1)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
VCC	Power supply voltage	-	-	3.3	-	V
VDCDC	DCDC outpu voltage	-	-	1.15	-	V
I_{load}	DCDC load current capability	Output current@ VDCDC=1.15V	-	-	20	mA

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η	DCDC convert rate(2)	-	-	82.5	-	%
V_{RPL}	DCDCT output voltage fluctuates	-	-	10	-	mV
L	DCDC Load inductance	-	1	2.2	10	uH
C_{OUT}	DCDC load capacitance	-	0.5	2	10	uF
t_{STAR}	DCDC output voltage settling time	-	-	90	-	uS

- Guaranteed by design, not tested in production.
- The inductor model used in the test is FDK MIPSDZ1608G2R2PA, and different inductors may cause differences in DCDC efficiency.

5.3.5. Current characteristics

Power consumption is a combination of parameters and factors, including operating voltage, ambient temperature, load on the I/O pins, software configuration of the product, operating frequency, flip rate of the I/O pins, location of the program in memory, and code execution. For a description of the current consumption measurement method, see Figure (Power Consumption Measurement Scheme).

Typical current consumption

Table 11 Typical current consumption in Sleep mode(1)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
I_{CC}	Current in Sleep	Low-speed clock on, 48K	-	1.6	3.8	uA

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	mode	SRAM keep, I/O status hold				
	Current in PD mode	VCC keep，WAKEUP IO and RESET can wake up system	-	0.13	1.0	uA

- Measurement conditions are TA =25°C、VCC=3.3V.

Typical current consumption in Run mode

The chip is working under the following conditions:

- All I/O pins are in a reset status
- All peripherals are turned off, unless specified
- Environment temperature and VCC supply voltage conditions are listed in the table (common operating conditions)

Table 12 Typical current consumption in Run mode

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
I_{CC}	Current supply in Run mode	Internal high-speed RC oscillator (HSI) (2)	-	2.0 (1)	-	mA

(1) Typical values are measured at TA = 25°C and VCC = 3.3V

(2) The internal high-speed clock is 64MHz

Table13 BLE power consumption

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
I_{CC}	Current	Transmit power 0dbm，VCC current	-	4.2	-	mA

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	supply in Run mode	Minimum sensitivity receiving, VCC current	-	3.8	-	mA
		1s broadcast interval，VCC average current	-	13	-	uA
		100ms broadcast interval，VCC average current	-	109	-	uA
		100ms connection interval, VCC average current	-	70	-	uA

- Typical values are measured at TA = 25°C and VCC = 3.3V

5.3.6. External clock source characteristics

A high-speed external clock is generated with a crystal/ceramic resonator.

The high-speed external clock (HSE) can be generated with a 32MHz crystal/ceramic resonator. The information given in this section is based on a comprehensive characterization using the typical external components listed in the table below. In applications, the resonator and load capacitors must be as close to the oscillator's pins as possible to reduce output distortion and settling time at start-up. For detailed parameters (frequency, package, accuracy, etc.) of crystal resonators, please consult the respective manufacturer. (The crystal resonator mentioned here is what we usually call a passive crystal oscillator)

Note: For CL1 and CL2, it is recommended to use a porcelain dicapacitor and pick a crystal or resonator that meets the requirements. Usually CL1 and CL2 have the same parameters.

Crystal manufacturers usually give the parameters of the load capacitance CL in a serial combination of CL1 and CL2, the load capacitance CL required for different crystals or resonators is usually different, and the choice of CL1 and CL2 must match the crystal or resonator used.

The load capacitance CL is calculated as follows: CL= CL1×CL2/(CL1+ CL2)+Cstray, where

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Cstray is the capacitance of the pin and the capacitance of the PCB board or PCB, and its typical value is between 2pF and 7pF.

Table 14 HSE 32MHz oscillator characteristic (1)(2)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
f_{OSC_IN}	Oscillator frequency	-	-	32	-	MHz
C_{L1} C_{L2}(3)	Recommended load capacitance	-	-	12(3)	-	pF
Rs(4)	Crystal Serial Impedance (R S)(4)	-	60	-	100	Ω
I_D	HSE drive current	VCC=3.3V, 12pF load	-	0.2	-	mA
$t_{SU(HSE)}$(5)	Start-up time		-	0.2	-	mS

(1) The characteristic parameters of the resonator are given by the crystal/ceramic resonator manufacturer

(2) Derived from laboratory testing, not tested in production

(3) For CL1 and CL2, it is recommended to use high-quality ceramic dielectric capacitors designed for RF applications and to select a crystal or resonator that meets the requirements. CL1 and CL2 are calculated from a given load capacitance from the crystal manufacturer and then adjusted according to the RF frequency characteristics. When selecting CL1 and CL2, the capacitive reactance of the PCB and chip pins should be taken into account

(4) Please refer to Crystal vendor for RS value

(5) tSU(HSE) is the start-up time, which is the time from the time the HSE is enabled by the software to the time when a stable 32MHz oscillation is obtained. This value is

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measured on a standard crystal resonator and can vary greatly depending on the crystal manufacturer.

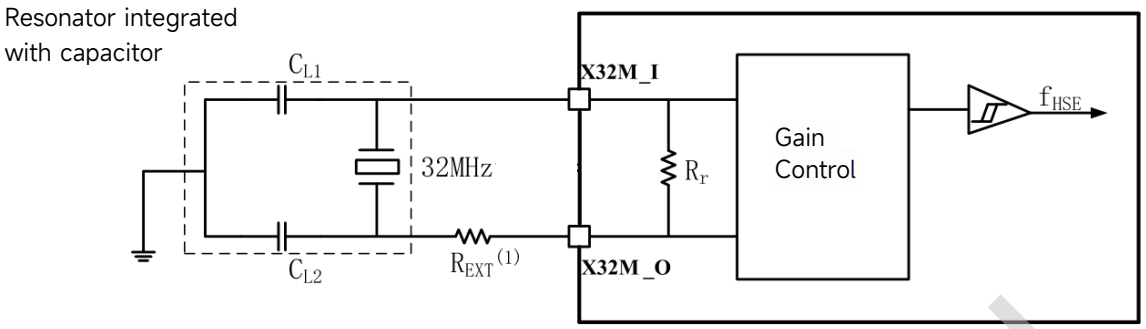


Figure 8 Typical applications using 32MHz crystals

(6) REXT value is determined by crystal characteristics.

A low-speed external clock generated by a crystal/ceramic resonator

A low-speed external clock (LSE) can be generated using an oscillator consisting of a 32.768kHz crystal/ceramic resonator. The information given in this section is based on a comprehensive characterization using the typical external components listed in the table below. In applications, the resonator and load capacitors must be as close to the oscillator's pins as possible to reduce output distortion and settling time at start-up. For detailed parameters (frequency, package, accuracy, etc.) of crystal resonators, please consult the respective manufacturer.

Note: For CL1 and CL2, it is recommended to use a porcelain dicapacitor between 8pF~20pF and pick a crystal or resonator that meets the requirements. Usually, CL1 and CL2 have the same parameters. Crystal manufacturers usually give the parameters of the load capacitance in a serial combination of CL1 and CL2, and the required load capacitance CL is usually different for different crystals or resonators, and the choice of CL1 and CL2 must match the crystal or resonator used.

The load capacitance CL is calculated by the following formula: CL = CL1×CL2 / (CL1 + CL2) + C stray where C stray is the capacitance of the pin and the capacitance of the PCB board or PCB correlation, which typically ranges from 2 pF to 7 pF.

Table 15 LSE oscillator characteristic (fLSE =32.768kHz) (1)

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Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
C_{L1} C_{L2}(2)	The recommended load capacitance and the corresponding crystal serial impedance (RS)(3)	RS：30KΩ~90KΩ	-	10	-	pF
I2	LSE drive current	VCC=3.3V，C L1 =C L2 =10Pf，RS=30KΩ	-	0.2	-	uA
$t_{SU(LSE)}$(4)	Start-up time		-	0.84	-	S

(1) Derived from laboratory testing, not tested in production.

(2) See the note paragraph at the top of this table

(3) Choosing a high-quality oscillator with a small RS value optimizes current consumption. Please consult the crystal manufacturer for details

(4) tSU(LSE) is the startup time, which is the time from the time the software enables LSE to obtain a stable oscillation of 32.768KHz. This value is measured on a standard crystal resonator and can vary greatly depending on the crystal manufacturer

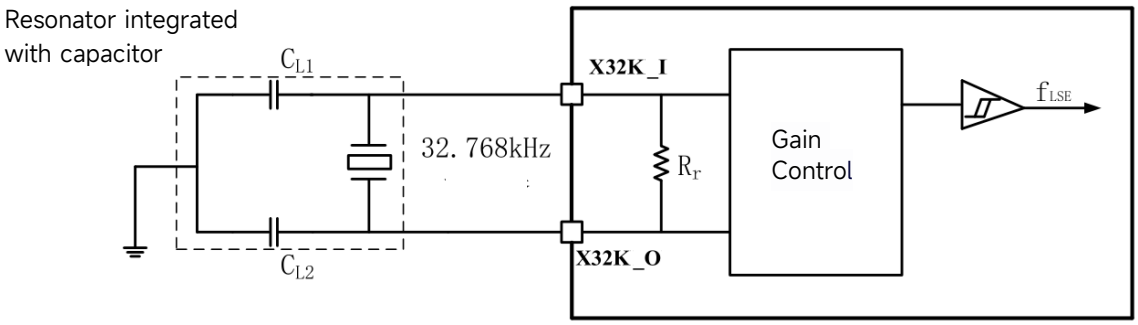


Figure 9 External low-speed clock

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5.3.7. Internal clock characteristic

The characteristic parameters given in the table below are measured at ambient temperature and supply voltage in Table 7 (General Operating Conditions).

High-speed internal RC oscillator

Table 16 HSI oscillator characteristic (1)(2)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
f_{HSI}	Frequency	T A = 25°C	63.36	64	64.64	MHz
ACC_{HSI}	HSI temperature drift of the oscillator	T A = -40~105°C, temperature drift	-3	-	3	%
		T A = -10~85°C, temperature drift	-2	-	2	%
		T A = 0~70°C, temperature drift	-1	-	1	%
$t_{SU(HSI)}$	HSI start-up time of oscillator	-	-	-	0.3	uS
$I_{CC(HSI)}$	HSI power consumption of oscillator	-	-	180	260	uA

(1) VCC = 3.3V, T A = -40~105°C.

(2) Guaranteed by design, not tested in production.

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Low-speed internal (LSI) RC oscillator

Table 17 LSI oscillator characteristic (1)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
f_{LSI}(2)	Output frequency	25°C after calibration	31.9	32	32.2	KHz
		T_A = -40~85°C, temperature drift	-1	-	1	%
$t_{SU(LSI)}$(3)	LSI start-up time of oscillator	-	-	-	200	uS
$I_{CC(LSI)}$(3)	LSI power consumption of oscillator	-	-	0.23	-	uA

(1) VCC = 3.3V, TA = -40~85°C.

(2) Derived from laboratory testing, not tested in production

(3) Guaranteed by design, not tested in production

5.3.8. Wake-up time from low power mode

The wake-up times listed in the table below are measured during the wake-up phase of a 64MHz HSI RC oscillator. The clock source used to wake up depends on the current mode of operation:

Sleep or PD mode: The clock source is an RC oscillator

All times are measured at ambient temperature and supply voltage in accordance with Table 7 (General Operating Conditions).

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Table 18 Wake-up time from low power mode

Symbol	Parameter	Condition	Min.	Typical	Max.
$t_{WUSLEEP}$(1)	Wake-up from Sleep mode	-	0.2	-	mS
t_{WUPD}(1)	Wake-up from PD mode	-	42	-	mS

(1) Wake-up time is the time between the start of the wake up event and the first instruction read by the user program

5.3.9. FLASH memory characteristics

Unless otherwise specified, all characteristic parameters are obtained at T A = -40~85°C.

Table 19Flash memory characteristic

Symbol	Parameter	Condition	Min.	Typical(1)	Max(1)	Unit
t_{PP}	Page (256 byte) program time	T_A = -40~85°C	-	2	3	mS
t_{PE}	Page (256 byte) erase time	T_A = -40~85°C	-	16	30	mS
t_{SE}	Sector (4K byte) erase	T_A = -40~85°C	-	16	30	mS
t_{CE}	Entire erase time	T_A = -40~85°C	-	16	30	mS

- (1) guaranteed by design, not tested in production.

Table 20 Flash memory lifetime and data retention period

Symbol	Parameter	Condition	Min.(1)	Unit

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N END	Life(Note：the number of erasures)	T_A = -40~85°C	10	10K
		T_A=-40~105°C	1	10K
t RET	Data retention period	T_A=105°C	20	Year

(1) Derived from laboratory testing, not tested in production.

5.3.10. Absolute maximum values(electrical sensitivity)

Based on different tests (ESD, LU), a chip is tested for strength with specific measurement methods to determine its performance in terms of electrical sensitivity.

ESD

The electrostatic discharge test (a positive pulse followed by a negative pulse after an interval of one second) is applied to all pins of all samples, and the size of the sample is related to the number of power pins on the chip (3 x (n+1) power pins). This test complies with MIL-STD-883K, ESDA/JEDEC JS-002-2018 standards.

Table 21 ESD absolute maximum value

Symbol	Parameter	Condition	Type	Min.(1)	Unit
$V_{ESD(HBM)}$	Electrostatic discharge voltage (manikin)	T_A = +25°C, comply with MIL-STD-883K	II	2000	V
$V_{ESD(CDM)}$	Electrostatic discharge voltage (charging equipment model)	T_A = +25°C, Comply with ESDA/JEDEC JS-002-2018	II	1000	V

(1) Derived from laboratory testing, not tested in production

Static latching

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For each power pin, an over-the-limit supply voltage is supplied.
Current is injected at each input, output, and configurable I/O pin.
This test complies with the JEDEC78E IC bolt lock standard.

Table 22 Electric sensitivity

Symbol	Parameter	Condition	Type
LU	Static latching	T_A = +25/+85°C，comply with JEDEC78E	II type A

5.3.11. I/O characteristics

GPIO characteristics

Unless otherwise specified, the parameters listed in the table below are measured under the conditions of Table 7 (General Operating Conditions). All I/O ports are CMOS and TTL compatible.

Table 23 I/O static characteristic(1)(2)

Symbol	Parameter	Condition	Min.	Max.	Unit
V_{IL}	Input low level voltage	VCC=3.3V	VSS	0.8	V
		VCC=2.5V	VSS	0.7	V
V_{IH}	Input high level voltage	VCC=3.3V	2	VCC	V
		VCC=2.5V	1.7	VCC	V
V_{hys}	Schmitt trigger voltage hysteresis (1)	VCC=3.3V/2.5V	200	-	mV

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I_{lkg}	Leakage current of input(3)	V_{PAD} =0 V_{PAD} =VCC	-1	1	uA
R_{PU}	Weak pull-up equivalent resistance (4)	VCC=3.3V V_{IN} = V_{IL}	120	140	KΩ
R_{PD}	Weak pull-down equivalent resistance (4)	VCC=3.3V V_{IN} = V_{IL}	120	140	KΩ
C_{IO}	I/O capacitance of the pins		-	0.1	pF

(1) (2) The hysteresis voltage at the switching level of the Schmitt trigger. Derived from laboratory testing, not tested in production

(3) If there is a reverse current backflush on the adjacent pins, the leakage current may be higher than the maximum

Output drive current

The GPIO can input or output a maximum of +/-12mA current.

Output voltage

Table 24 I/O output voltage

Symbol	Parameter	Condition	Min.	Max.	Unit
V_{OL}	Output low level voltage	VCC=3.3, I_{OH} =2mA, 4mA, 8mA, 12mA	VSS	0.4	V
		VCC=2.5, I_{OH} =2mA, 4mA, 8mA, 12mA	VSS	0.4	V

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V_{OH}	Output high level voltage	VCC =3.3V, I_{OH} = -2mA, -4mA, -8mA, -12mA	2.4	VCC	V
		VCC =2.5V, I_{OH} = -2mA, -4mA, -8mA, -12mA	2	VCC	V

Input/output AC characteristics

The definitions and values of the input/output AC characteristics are given in the table below. Unless otherwise specified, parameters are measured at ambient temperature and supply voltage in Table 7 (General Operating Conditions).

Table 25 Input/output AC characteristics(1)

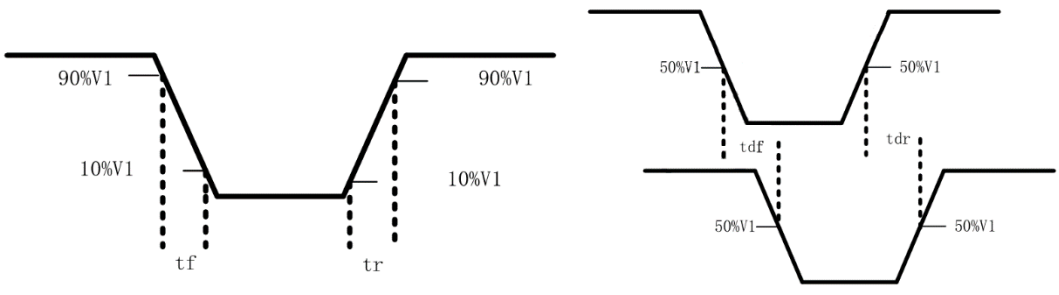
Register configuration	Symbol	Parameter	Condition	Min.	Max.	Unit
00 (2mA)	$f_{max(IO)out}$	Maximum frequency	C_L=5pF, VCC =3.3V	-	64	MHz
			C_L=5pF, VCC =2.5V	-	50	MHz
	$t_{(IO)out}$	Output delay	C_L=5pF, VCC =3.3V	-	3.66	nS
			C_L=5pF, VCC =2.5V	-	4.72	
	$t_{(IO)in}$	Input delay	C_L=50fF, VCC =2.97V VCC= 2.5V	-	1.2	nS
01 (4mA)	$f_{max(IO)out}$	Maximum frequency	C_L=10pF, VCC =3.3V	-	64	MHz
			C_L=10pF, VCC =2.5V	-	60	MHz

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	$t_{(IO)out}$	Output delay	C_L=10pF，VCC =3.3V	-	3.5	nS
			C_L=10pF，VCC =2.5V	-	4.5	nS
	$t_{(IO)in}$	Input delay	C_L=50fF，VCC =2.97V	-	1.2	nS
			C_L=50fF，VCC =2.5V	-		
10 (8mA)	$f_{max(IO)out}$	Maximum frequency	C_L=20pF，VCC =3.3V	-	64	MHz
			C_L=20pF，VCC =2.5V	-	50	MHz
	$t_{(IO)out}$	Output delay	C_L=20pF，VCC =3.3V	-	3.42	nS
			C_L=20pF，VCC =2.5V	-	4.73	nS
	$t_{(IO)in}$	Input delay	C_L=50fF，VCC =2.97V	-	1.2	nS
			C_L=50fF，VCC =2.5V	-		
11 (12mA)	$f_{max(IO)out}$	Maximum frequency	C_L=30pF，VCC =3.3V	-	64	MHz
			C_L=30pF，VCC =2.5V	-	50	MHz
	$t_{(IO)out}$	Output delay	C_L=30pF，VCC =3.3V	-	3.34	nS
			C_L=30pF，VCC =2.5V	-	4.26	nS
	$t_{(IO)in}$	Input delay	C_L=50fF，VCC =2.97V	-	1.2	nS
			C_L=50fF，VCC =2.5V	-		

(1) Guaranteed by design, not tested in production.



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Figure 10 Input/output AC characteristic definition

5.3.12. RESET PIN characteristics

The RESET pin input driver uses CMOS technique. It is connected to a pull-up resistor RPU that cannot be disconnected. Unless otherwise specified, parameters listed in the table below are measured under the conditions of ambient temperature and supply voltage in Table 7 (General Operating Conditions).

Table 26 RESET pin characteristics

Symbol	Parameter	Condition	Min.	Max.	Unit	Symbol
$V_{IL(RESET)}$ (1)	RESET input low level voltage	VCC = 3.3V	VSS	-	0.8	V
$V_{IH(RESET)}$ (1)	RESET input high level voltage	VCC = 3.3V	2	-	VCC	V
$V_{hys(RESET)}$ (1)	RESET Schmitt trigger voltage hysteresis	-	-	200	-	mV
R_{PU}	Weak pull-up equivalent resistance	VCC = 3.3V	40	50	60	KΩ
$V_{F(RESET)}$ (1)	RESET input filter pulse	-	-	-	500	ns
$V_{NF(RESET)}$ (1)	RESET input non-filter pulse	-	5	-	-	us

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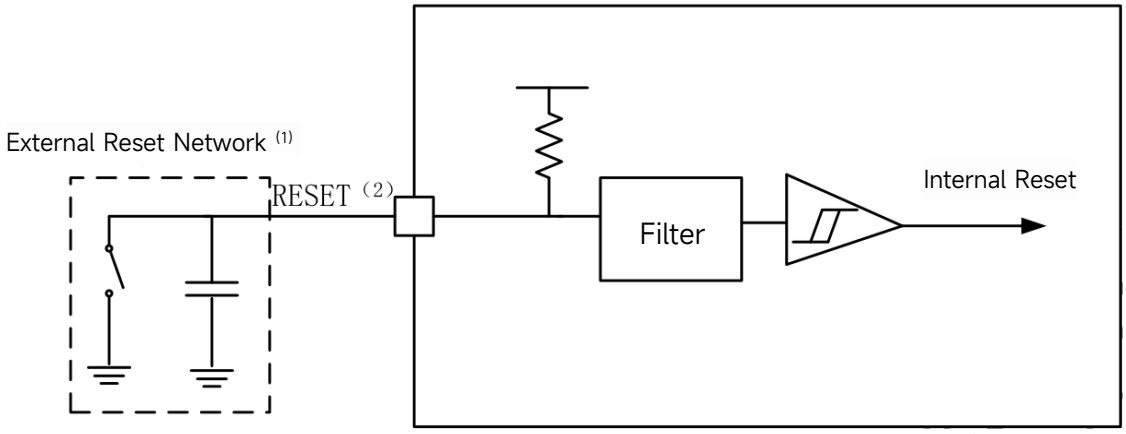


Figure 11 Recommended RESET pin protection

(1) Guaranteed by design, not measured in production

- The external reset network is to prevent parasitic reset

(2) The user must ensure that the potential of the RESET pin is below the maximum VIL (RESET) listed in the table above, otherwise the chip cannot be reset。

5.3.13. TIM timer characteristics

The parameters listed in the table below are measured at ambient temperature and supply voltage in Table 7 (General Operating Conditions).

Table 27 Characteristics of TIMx(1)(2)

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Symbol	Parameter	Condition	Min.	Max.	Unit
$t_{res(TIM)}$	Timer resolution time	$f_{TIMxCLK}$ = 64MHz	1	-	$t_{TIMxCLK}$
			15.625	-	ns
f_{EXT}	CH1~CH4 timer external clock frequency		0	$f_{TIMxCLK}/2$	MHz
			0	32	MHz
Res_{TIM}	Timer resolution		-	16	位
$t_{COUNTER}$	When an internal clock is selected, the 16-bit counter clock cycles		1	65536	$t_{TIMxCLK}$
			0.015625	1024	us
t_{MAX_COUNT}	The maximum possible count		-	65536x65536	$t_{TIMxCLK}$
			-	67.1	s

(1) TIMx is a generic name which represents TIM1/TIM3/TIM6.

(2) Parameters are guaranteed by the design.

5.3.14. I2C interface characteristics

Unless otherwise specified, parameters are measured in Table 7 (General Operating Conditions) at ambient temperature, fPCLK1 frequency, and VCC supply voltage in accordance with Table conditions.

The I2C interface conforms to the standard I2C communication protocol, but has the following limitations: SDA and SCL are not "true" open-drain pins, and when configured as an open-drain output, the PMOS tube between the pinout pin and VCC is turned off, but still exists.

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The I2C interface characteristics are shown in the table below, and for more information on the characteristics of the input/output multiplexing function pins (SDA and SCL), see (I/O Port Characteristics).

Table 28 I2C interface characteristics(1)

Symbol	Parameter	Standard mode		Fast mode		Unit
		Min.	Max.	Min.	Max.	
f_{SCL}	I2C interface frequency	-	100	-	1000	KHz
$t_{h(STA)}$	Hold time of start-up condition	4.0	-	0.6	-	us
$t_{w(SCLL)}$	SCL clock low time	4.7	-	1.3	-	us
$t_{w(SCLH)}$	SCL clock high time	4.0	-	0.6	-	us
$t_{su(STA)}$	The start condition establishment time of the repetition	4.7	-	0.6	-	us
$t_{h(SDA)}$	SDA data hold time	-	3.4	-	0.9	us
$t_{su(SDA)}$	SDA data establishment time	250		100		ns
$t_{r(SDA)}$ $t_{r(SCL)}$	SDA and SCL rise time	-	1000	20	300	ns
$t_{f(SDA)}$ $t_{f(SCL)}$	SDA and SCL fall time	-	300	-	300	ns
$t_{su(STO)}$	Stop condition establishment time	4.0	-	0.6	-	us
$t_{w(STO:STA)}$	The time from the stop condition to	4.7	-	1.3	-	us

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	the start condition (bus idle)					
C_b	Capacitive load on each bus	-	400	-	100	pF
$t_{v(SDA)}$	Valid time of Data	3.45	-	0.9	-	us
$t_{v(ACK)}$	Valid time of acknowledge	3.45	-	0.9	-	us

(1) Guaranteed by the design, not measured in production.

(2) To achieve the maximum frequency of standard mode I2C, fPCLK1 must be greater than 2MHz. In order to achieve the maximum frequency of fast mode I2C, fPCLK1 must be greater than 10MHz.

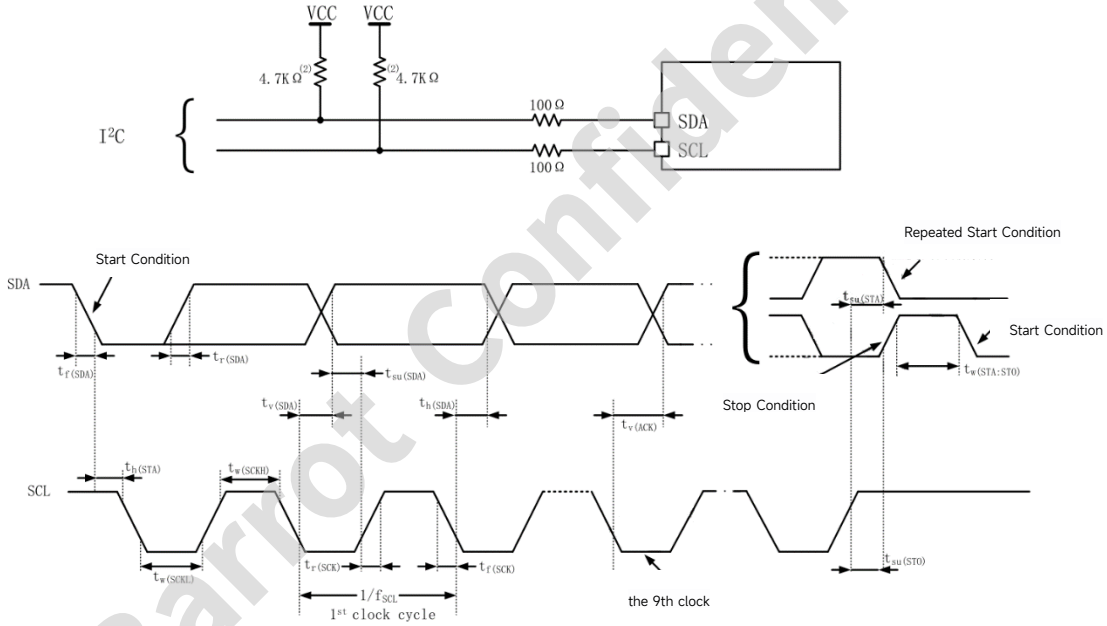


Figure 12 I2C Bus AC waveforms and measurement circuits (1)

(1) The measurement points are set at CMOS levels: 0.3VCC and 0.7VCC.

5.3.15. SP interface characteristics

Unless otherwise specified, SPI parameters are measured in Table 7 (General Operating Conditions) at ambient temperature, fPCLK2 frequency, and VCC supply voltage in accordance with Table conditions.

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For details on the characteristics of the input/output multiplexing function pins (NSS, SCLK, MOSI, MISO), see (I/O Port Characteristics).

Table 29 SPI characteristics(1)

Symbol	Parameter	Condition		Min.	Max.	Unit
f_{SCLK} $1/t_{c(SCLK)}$	SPI clock frequency	Master mode		-	16	MHz
		Slave mode		-	16	
$t_{r(SCLK)}$ $t_{f(SCLK)}$	SPI clock rising and fall time	Load capacitance: C = 30pF		-	8	us
$DuCy_{(SCK)}$	SPI slave input clock duty cycle	SPI slave mode		30	70	%
$t_{su(NSS)}$(1)	NSS establishment time	Slave mode		4_{tPCLK}	-	ns
$t_{h(NSS)}$(1)	NSS keep time	Slave mode		2_{tPCLK}	-	ns
$t_{w(SCLKH)}$(1) $t_{w(SCLKL)}$(1)	SCLK high and low time	Master mode		t_{PCLK}-2	t_{PCLK}+2	ns
$t_{su(MI)}$(1)	Data input establishment time	Master mode	SPI1	5	-	ns
			SPI2	6	-	
$t_{su(SI)}$(1)		Slave mode	SPI1	5	-	ns
			SPI2	6	-	
$t_{h(MI)}$(1)	Data input hold time	Master mode		4	-	ns
$t_{h(SI)}$(1)		Slave mode		3	-	ns

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$t_{a(SO)}$(1)(2)	Data output access time	Slave mode, f_{PCLK} = 32MHz		0	3_{tPCLK}	ns
$t_{dis(SO)}$(1)(3)	Disable time of Data output	Slave mode		2	10	ns
$t_{v(SO)}$(1)	Valid time of data output	Master mode	SPI1	-	16	ns
			SPI2	-	20	
$t_{v(MO)}$(1)		Slave mode	SPI1	-	8	
			SPI2	-	10	
$t_{h(SO)}$(1)	Data output hold time	Slave mode (after edge of enabling)		2	-	ns
$t_{h(MO)}$(1)		Master mode (after edge of enabling)		0	-	

(1) Derived from the test in laboratory, not measured in production.

(2) The minimum value represents the minimum time to drive the output, and the maximum value represents the maximum time to obtain the data correctly

(3) The minimum value represents the minimum time to turn off the output, and the maximum value represents the maximum time to put the data line in a high-impedance state.

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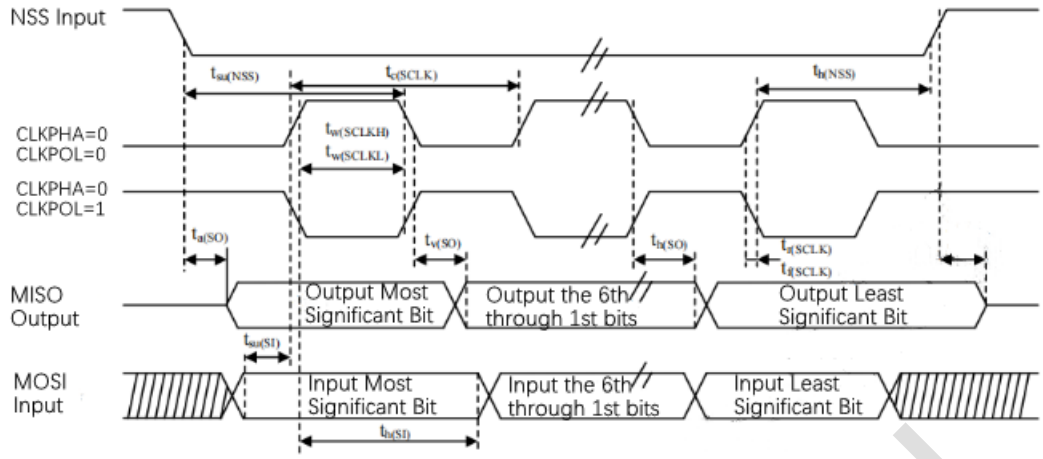


Figure 13 SPI timing diagram – from mode and CPHA=0

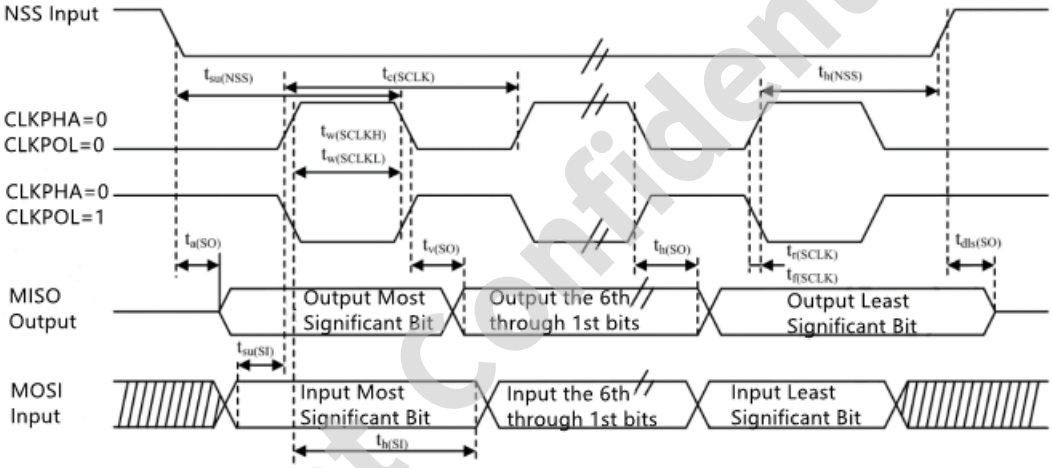


Figure 14 SPI Timing Diagram – Slave Mode and CPHA =1(1)

(1) The measurement points are set at CMOS levels: 0.3VCC and 0.7VCC.

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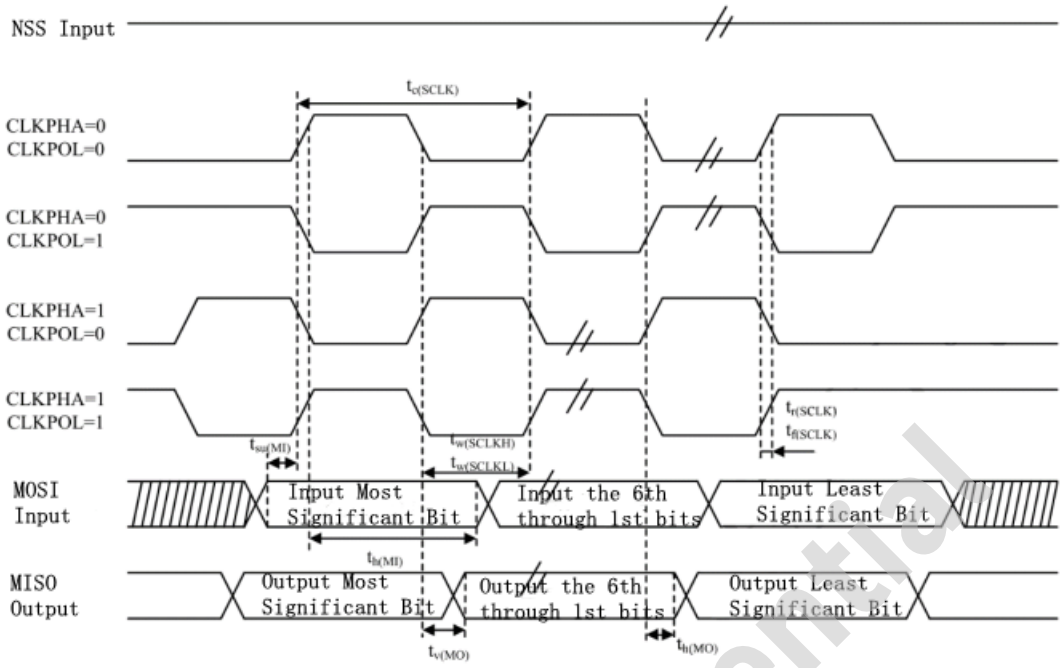


Figure 15 SPI timing diagram – Main Mode (1)

(1) The measurement points are set at CMOS levels: 0.3VCC and 0.7VCC.

5.3.16. Temperature sensor characteristics

Unless otherwise specified, parameters are measured in Table 7 (General Operating Conditions) at ambient temperature, fHCLK frequency and VCC supply voltage in accordance with Table conditions.

Table 30 Temperature sensor characteristics

Symbol	Parameter	Condition	Min.	Max.	Unit
T_L(1)	V_{SENSE} linearity with respect to temperature	-	±4	-	°C
Avg_Slope(1)	Average slope	-	2.14(2)	-	mV/°C
t_{START}(1)	Establishment time	-	-	10	us

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(1) Guaranteed by design, not tested in production

(2) Derived from laboratory testing, not tested in production

5.3.17. ADC characteristics

Unless otherwise specified, parameters are measured in Table 7 (General Operating Conditions) at ambient temperature, fHCLK frequency and VCC supply voltage in accordance with Table conditions.

Table 31 ADC characteristics

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
V_{REF+}	Positive reference voltage	-	-	1.0	-	V
f_{ADC}	ADC sampling speed	-	-	-	1.33	MHz
V_{AIN}	Conversion voltage range, external low-voltage path	-	0	-	1000	mV
	Conversion voltage range, external high-voltage path	-	0	-	3600(2)	mV
SNDR	Singal-to-Noise-a	Input frequency	-	46	-	dBFS

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	nd-Distortion Ratio Power-up time	=1.03KHz, VCC=3.3V, T A =25°C f_{ADC} =1Msps				
		-	-	58	-	
t_{STAB}(1)	Convert time	-	-	16	-	us
t_{CONV}(1)	Differential linearity error	VCC=3.3V, T A =25°C	752	-	-	ns
DNL	Integral linearity error	VCC=3.3V, T A =25°C	-1	-	6	LSB
INL	Singal-to-Noise-and-Distortion Ratio	Input frequency =1.03KHz, VCC=3.3V, T A =25°C f_{ADC} =1Msps	-8	-	2	LSB

(1) Derived from laboratory testing, not tested in production

(2) The maximum value is 3600mV and < = VCC+300mV

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5.3.18. PGA characteristics

Unless otherwise specified, parameters are measured in Table 7 (General Operating Conditions) at ambient temperature, and VCC supply voltage in accordance with Table conditions.

Table 32 PGA characteristics

Symbol	Parameter	Min.	Typical	Max.	Unit

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GAIN	PGA gain	0	-	42	dB
GAIN STEP	PGA gain step	-	6	-	dB
THD+N(1)	Gain=0dB	73	82	85	dB
	Gain=42dB	73	83	87	dB
In-band ripple(1)	300~3400Hz gain fluctuations within the frequency band	-	0.78	-	dB
T_{PGA}(1)	PGA establishment time	-	15	-	ms
MIC_BIAS voltage	MIC bias voltage，step=0.1v	1.6	-	2.3	V
MIC_BIAS Noise(1)	20Hz to 8kHz A-weighted with 4.7uF	-	-92	-	dBV

(1) Derived from laboratory testing, not tested in production.

5.3.19. KEYSCAN characteristics

Unless otherwise specified, parameters are measured in Table 7 (General Operating Conditions) at ambient temperature, and VCC supply voltage in accordance with Table conditions.

Table 33 KEYSCAN characteristics

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
T_{WTS}	Time gap for each round of keyboard scanning	-	-	32	224	ms

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T_{DTS}	Key debounce time	-	10	-	640	ms
I_{cc}(1)	Power consumption in auto-scan mode (104 keys)	T_{DTS}=40ms, T_{WTS} =32ms VCC=3.3V, T_A =25°C	-	2.9	-	uA
	Low power mode power consumption (104 buttons)	VCC=3.3V, TA =25°C	-	2.3	-	uA

(1) Derived from laboratory testing, not tested in production.

5.3.20. BLE characteristics

Unless otherwise specified, parameters are measured in Table 7 (General Operating Conditions) at ambient temperature, and VCC supply voltage in accordance with Table conditions.

Table 34 BLE receiver characteristics(1)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
1	Sensitivity, 1Mbps	VCC=3.3V, T_A =25°C	-	-96	-	dBm
2	Sensitivity, 2Mbps		-	-93	-	dBm
3	Co-channel interference		-	8		dB
4	Proximity channel interference, +-1MHz		-	1	-	dB

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5	Proximity channel interference，+-2MHz		-	-31	-	dB
6	Proximity channel interference，>=+-3MHz		-	-40	-	dB
7	Mirror channel interference		-	-24	-	dB
8	Proximity channel interference，+-1MHz		-	-28	-	dB
9	Maximum input power		-	-	6	dBm

(1) Derived from laboratory testing, not tested in production.

Table 35 BLE transmitter characteristics(1)

Symbol	Parameter	Condition	Min.	Typical	Max.	Unit
1	Output power	VCC=3.3V，T_A =25°C	-	-	6	dBm
2	Frequency accuracy		-	7.5	-	KHz
3	Frequency drift speed		-	-9.4	-	kHz/50us
4	Frequency drift		-	-15.1	-	KHz
5	Initial frequency drift		-	-13.2	-	KHz
6	Δf1 平均		-	258	-	KHz
7	Δf2 99.9%		-	218	-	KHz
8	Δf2/Δf1		-	1.06	-	-

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9	Harmonic power, second harmonic		-	-26	-	dBm
10	Harmonic power, third harmonic		-	-28	-	dBm
11	Harmonic power, fourth harmonic		-	-54	-	dBm
12	Harmonic power, fifth harmonic		-	-55	-	dBm

(1) Derived from laboratory testing, not tested in production.

6. Schematics and design

Please refer to《Schematics_BR1601A02.pdf》.

7. Package size

Package size: QFN32(4X4mm)

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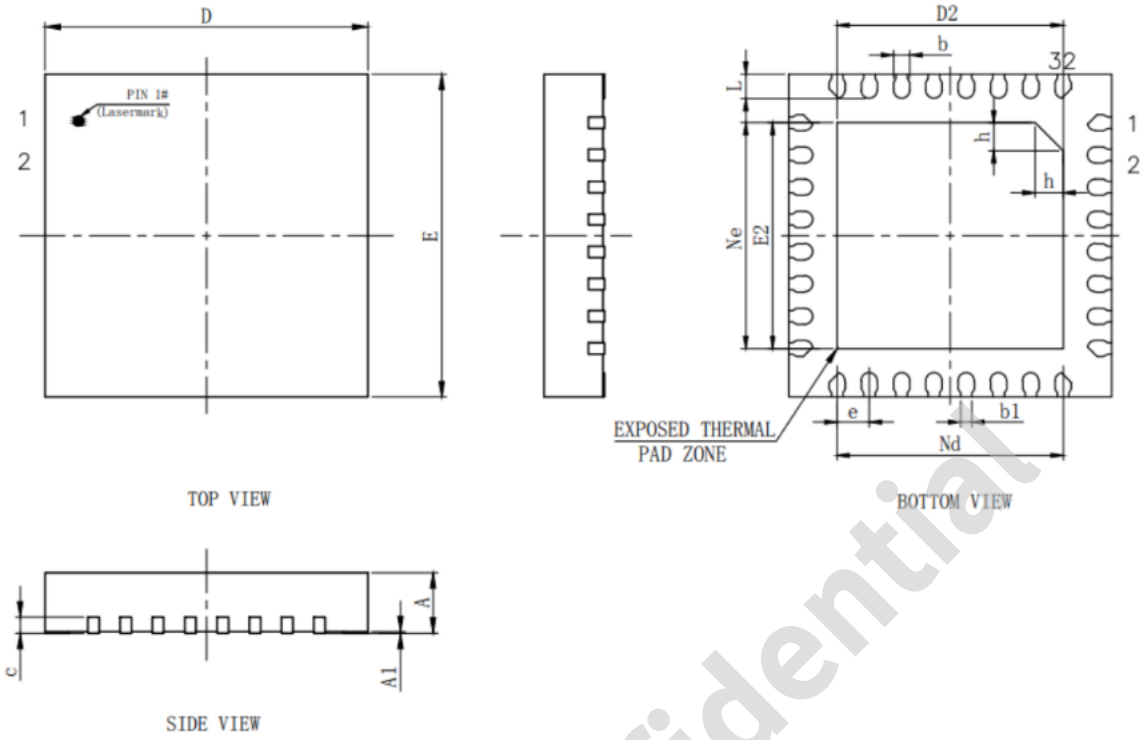


Figure 16 BR1601A02 package size

Table 36 Package size

Symbol	Millimeter		
	Min	Typical	Max
A	0.70	0.75	0.80
	0.80	0.85	0.90
	0.85	0.90	0.95
A1	0	0.02	0.05
b	0.15	0.20	0.25
b1	0.14REF		

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c	0.18	0.20	0.25
D	3.90	4.00	4.10
D2	2.70	2.80	2.90
e	0.40BSC		
Ne	2.80BSC		
Nd	2.80BSC		
E	3.90	4.00	4.10
E2	2.70	2.80	2.90
L	0.25	0.30	0.35
h	0.30	0.35	0.40
L/F	122X122		

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8. Package

Package: Reel

Minimum quantity per pack: 3000 PCS/reel

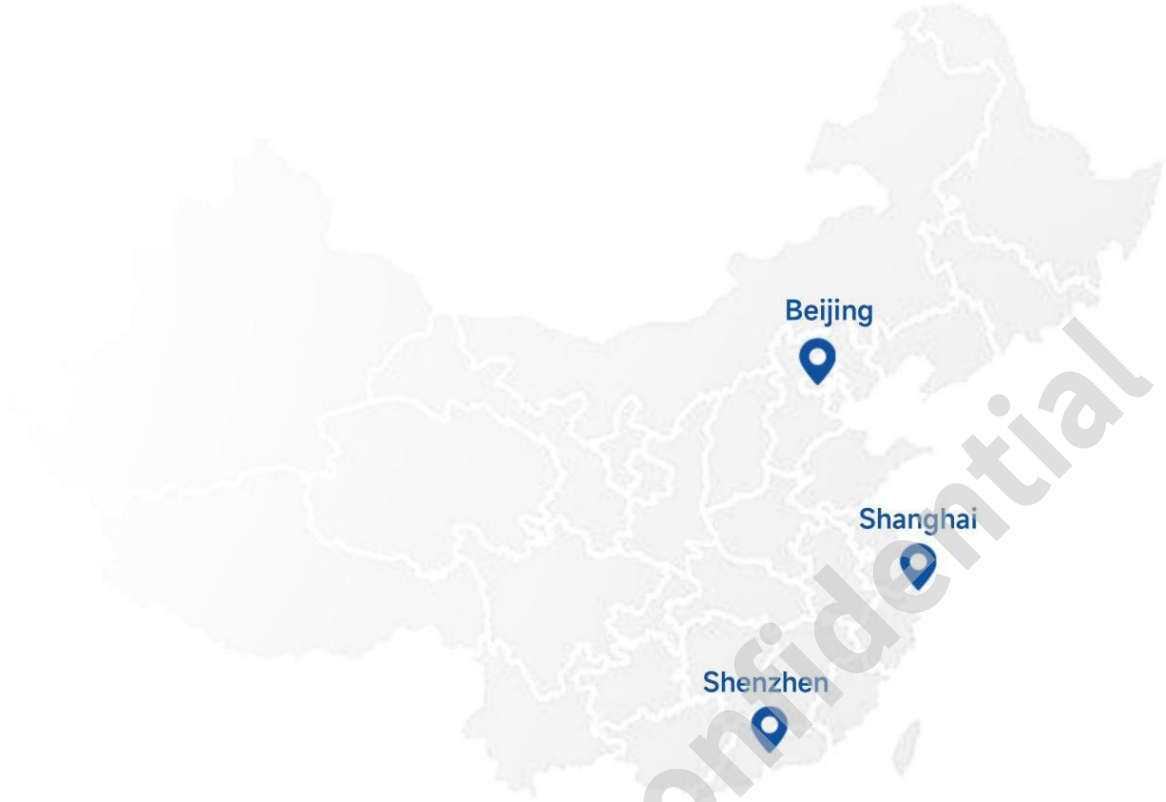
9. Company profile

Barrot - the world's leading one-stop chip-level wireless interconnection Internet of Things solution provider with independent intellectual property rights, is a senior member of Bluetooth International Organization (Bluetooth SIG), the only Bluetooth standard formulation company in China, with independent intellectual property rights of Bluetooth radio frequency, protocol stack, audio algorithm three major IP, focusing on the Internet of Things and automotive electronics.

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